

4. Project Background

Truck loading operations at oil terminals require strict adherence to a defined sequence: correct compartment selection, secure loading arm connection, grounding/bonding verification, and confirmation of vapor recovery connection (where applicable) prior to the release of product flow. Deviations from this sequence — whether from human error, time pressure, or communication gaps between the driver and the terminal operator — are a recurring contributor to product loss, equipment damage, and Loss of Primary Containment (LOPC) events across the industry.

Site data analysis at Gantry #15 identifies loading time as the main contributing bottleneck, accounting for approximately 49% of total truck turnaround time. Much of this time is attributable to conservative, sequential single-arm loading practice (one compartment/arm at a time to minimize risk), rather than to a technical constraint — indicating a clear opportunity for AI-advised loading pattern optimization to reduce loading time variation without changing the underlying safety procedure.

The Customer currently relies on manual visual supervision by loading bay operators and drivers, supported by interlocks within the existing Terminal Automation System (TAS) for critical safety functions (e.g., overfill protection, earthing continuity). However, there is currently no automated, camera-based visual verification layer that independently confirms loading arm engagement and compartment alignment before and during product transfer.

The Customer has already invested in an ENIM AI Server platform as part of a broader digitalization initiative, together with supporting fiber-optic and LAN network infrastructure. This proposal is scoped specifically to extend the use of that existing AI compute investment into a new, high-value use case — visual loading verification — rather than proposing a parallel, standalone AI platform.

Business Drivers

- Reduce product loss and demurrage associated with loading sequence errors and rework.
- Strengthen HSE performance by adding an independent, camera-based verification layer to existing procedural and interlock-based controls.
- Build an in-house, terminal-specific AI dataset and capability, rather than depend solely on off-the-shelf analytics.
- Establish a low-risk, single-bay PoC that provides a clear, measurable basis for a terminal-wide investment decision (Phase II).

5. Current System Overview

This section summarizes the existing terminal infrastructure that the Smart Load Vision System will interface with. All items below are assumed to be Customer-owned, commissioned, and available for interconnection; verification of exact make/model, spare capacity, and physical condition will be performed during the Detailed Engineering phase (Section 11).

Existing Asset	Current Function	Relevance to This Project
ENIM AI Server	Central AI compute platform for existing digitalization use cases	Hosts the AI inference workload for loading arm / compartment detection (Section 9)
Fiber Optic Backbone	Terminal-wide data communication between field and control room	Carries camera and HMI network traffic to the server room (Section 13)
LAN Network	Existing IT/OT data network within control building / server room	Carries camera, HMI, dashboard, and database traffic on the existing terminal LAN
24VDC Power Supply (in JB)	Powers existing field instrumentation at the loading bay	Provides regulated 24VDC source for the new Smart HMI and the LINK PS-3180 Managed Industrial PoE+ Switch, subject to spare capacity check (Section 14)
Explosion Proof Junction Box	Local termination point for field cabling in the hazardous area	Reused as the local termination and cable-entry point for new camera and HMI circuits
Terminal Automation System (TAS)	Loading arm control, overfill protection, sequence interlocks	Acts as a read-only Data Provider to the new AI system via TAS API (HTTPS / REST API); no write-back or control command (Section 13)

Assumptions on Existing Infrastructure

- The existing 24VDC power supply and existing fiber/LAN infrastructure have adequate spare capacity for the additional Phase I load; this is to be confirmed by field survey and is treated as a key assumption of this proposal (see Section 18 — Risk Assessment).
- The existing explosion proof junction box has at least two spare, appropriately rated cable-entry positions, or the Customer will approve installation of a supplementary Ex e junction box adjacent to the existing one.
- The ENIM AI Server has sufficient available GPU/CPU headroom to run the proposed inference workload without impacting existing applications; a resource allocation review is included in Detailed Engineering.
- TAS integration in Phase I is limited to read-only data consumption from the existing TAS via TAS API (HTTPS / REST API) — no write-back, no control command, no modification to existing TAS control logic, safety logic, or SIL-rated functions is included or required in this phase.
- TAS API specification and access credentials shall be provided by the Customer; the exact data items available are subject to confirmation during Detailed Engineering (Section 12.5 — TAS API Data Interface).
- Site access, work permit, shutdown window, and installation approval shall be provided by the Customer; the final hazardous area classification shall be confirmed by the Customer's hazardous area classification (HAC) drawing.

6. Project Objectives

The Phase I Proof of Concept is designed around the following primary and secondary objectives, which also form the basis of the acceptance criteria defined in Section 15 (Testing and Commissioning).

Primary Objectives

1. Demonstrate reliable, real-time AI detection of loading arm connection status (connected / not connected / partially connected) with a target accuracy of $\geq 95\%$ under normal daytime and nighttime lighting conditions.
2. Demonstrate reliable identification of truck compartments within the camera field of view, sufficient to support compartment-sequence verification logic.
3. Provide real-time advisory feedback to the loading bay operator through the explosion-proof Smart HMI, with a target end-to-end latency of less than 1.5 seconds from event occurrence to HMI display.
4. Establish a structured event and video-clip database to support both operational review (incident investigation) and AI model retraining.

Secondary Objectives

5. Validate the mechanical, electrical, and network integration approach for a hazardous-area, camera-based AI system using the Customer's existing infrastructure, to de-risk Phase II terminal-wide deployment.
6. Establish an advisory-only (non-intrusive) integration pattern with the existing TAS that can be extended, in a future phase and subject to separate MOC (Management of Change) and SIL review, toward supervised closed-loop functions.
7. Build internal Customer familiarity and operator trust in AI-assisted loading supervision ahead of wider rollout.
8. Quantify measurable KPIs (loading time, exception rate, near-miss detection) to support the Phase II business case.

7. Challenges

The following technical and operational challenges have been specifically considered in the design of this proposal. Each is addressed by a corresponding design decision referenced in the section indicated.

Challenge	Design Response	Reference
Camera and HMI must operate reliably within the applicable hazardous area classification at Gantry #15, as confirmed by the Customer's HAC drawing: camera location Zone 2 / Group IIB / T5 and HMI location Zone 1 / Group IIB / T3.	Selection of factory-certified Ex-rated camera (Dahua IPC-HFE8849F-Z-PV) and Ex-rated Smart HMI (BARTEC POLARIS); Ex-rated cable glands, conduit, and junction box practices throughout.	Sec. 10, 11
Variable lighting conditions (day / night / backlight from vessel lighting) affect detection accuracy.	8MP / 4K sensor with wide dynamic range and motorized zoom for optimal framing; AI model trained/tuned on-site during the calibration period (Sec. 17).	Sec. 10, 15
Reuse of existing infrastructure (server, network, power) without disrupting existing operational systems.	Non-intrusive use of the existing terminal LAN (no new network segmentation), dedicated resource allocation review on the ENIM AI Server, and load-verified reuse of the 24VDC supply.	Sec. 13, 14
Integration with TAS without introducing safety risk or affecting existing control functions.	Strictly read-only data consumption via TAS API (HTTPS / REST API); no write-back, no control command, no modification to existing control or safety logic.	Sec. 13
Limited historical labeled data for terminal-specific AI model training.	Phased approach: baseline model trained on vendor's industry dataset, followed by on-site calibration using terminal-specific footage collected during commissioning.	Sec. 12, 17
Operator acceptance and change management for a new advisory tool.	Structured operator training program (Sec. 16, 20) and a simple, uncluttered HMI advisory design to minimize workload.	Sec. 16, 20

8. Proposed Solution

The Smart Load Vision System is an integrated Computer Vision and AI advisory platform purpose-built for hazardous-area truck loading bays. It combines an AI-ready, explosion-proof IP camera (Dahua IPC-HFE8849F-Z-PV) with server-side pre-processing and inference on the Existing ENIM AI Server, an explosion-proof operator/driver advisory HMI, and a supervisory dashboard, layered on top of the Customer's existing ENIM AI Server, network, and power infrastructure. The following subsections describe each functional building block.

8.1 AI Detection and Computer Vision

The core of the system is a deep-learning object detection and state-classification pipeline. Video streams from the AI camera are transmitted to the Existing ENIM AI Server. Pre-processing, region-of-interest extraction, and frame enhancement are performed on the server, followed by the primary inference pass, which classifies: (a) loading arm position and connection state, (b) truck/trailer compartment boundaries and identifiers, (c) presence and correct positioning of grounding/bonding cable (where visible in frame), and (d) generic safety observations such as personnel presence in the restricted loading envelope. The AI camera itself performs no on-board AI inference in Phase I; it provides high-resolution video only.

8.2 Loading Pattern Recognition

Beyond single-frame detection, the Advisory Logic Engine (Section 12.6) applies temporal pattern recognition across the video stream to reconstruct the loading sequence as a state machine: Truck Positioned → Grounding Connected → Arm Connected → Compartment Confirmed → Loading in Progress → Arm Disconnected → Truck Cleared. Deviations from the expected state order — for example, product flow implied before arm-connected confirmation, or a compartment mismatch against the terminal's loading order — are flagged as exceptions in real time.

8.3 Driver / Operator Advisory

The Smart HMI shall provide advisory information to the loading operator and/or driver-facing operation area, depending on the final HMI installation position confirmed during site survey. Advisory messages are presented via the explosion-proof Smart HMI / Advisory Display using a simple traffic-light and short-text convention (e.g., GREEN — Sequence Normal, AMBER — Verify Connection, RED — Sequence Exception / Stop). Depending on the confirmed installation position, the unit may be referred to as an Operator Advisory HMI (mainly for the bay operator), a Driver Advisory Display (mainly visible to the driver), or a Smart HMI / Advisory Display where both audiences are applicable. The HMI is intentionally designed as an advisory and decision-support tool: it does not issue automatic stop/start commands to the loading arm in this phase; the operator retains full authority and existing TAS interlocks remain the sole safety-instrumented control layer.

8.4 Dashboard

A web-based supervisory dashboard, hosted on the existing server LAN, provides terminal supervisors and HSE personnel with a live view of loading bay status, a searchable event history with linked video clips, and summary KPI reporting (average loading time, exception rate by shift/operator/truck, and AI confidence trending). The dashboard is accessible from standard office workstations over the existing corporate network, subject to the access control policy defined in Section 13.5.

8.5 Event Logging

Every detected state transition, advisory message, and operator acknowledgment is time-stamped and written to a structured event database, together with a short (configurable, typically 10–15 second) video clip covering the event window. This creates a complete, auditable record for incident investigation, regulatory reporting, and — critically — a growing labeled dataset that is used to retrain and improve the AI model over the life of the PoC and into Phase II.

9. System Architecture

The system architecture is presented below in four complementary views: overall layered architecture, network architecture, data flow, and physical hardware connectivity. This multi-view presentation follows standard industrial automation documentation practice (comparable to a typical DCS/PAS vendor FEED package) to give both engineering and management reviewers a complete picture of the proposed solution.

9.1 Overall Architecture

Figure 9.1 presents the system as five logical layers, with a clear boundary drawn between the hazardous area (applicable gas/dust zones per the Customer's hazardous area classification) field equipment and the safe-area compute and presentation equipment. This layered view is used consistently throughout this proposal (equipment specification, installation scope, and testing sections all map back to these five layers). The field camera sends video only — all pre-processing, AI inference, advisory logic, Smart HMI / dashboard processing, and event logging / database functions are performed on the Existing ENIM AI Server. The Existing TAS provides data to the Existing ENIM AI Server through TAS API (HTTPS / REST API); no AI write-back or control command is sent to TAS. The network backbone layer reflects the confirmed final design: a LINK PS-3180 Managed Industrial PoE+ Switch in the field, providing field LAN and PoE to the camera, with the existing fiber reused and uplinked to a LINK UT-1310A Gigabit Media Converter at the Control Room side (Section 13).

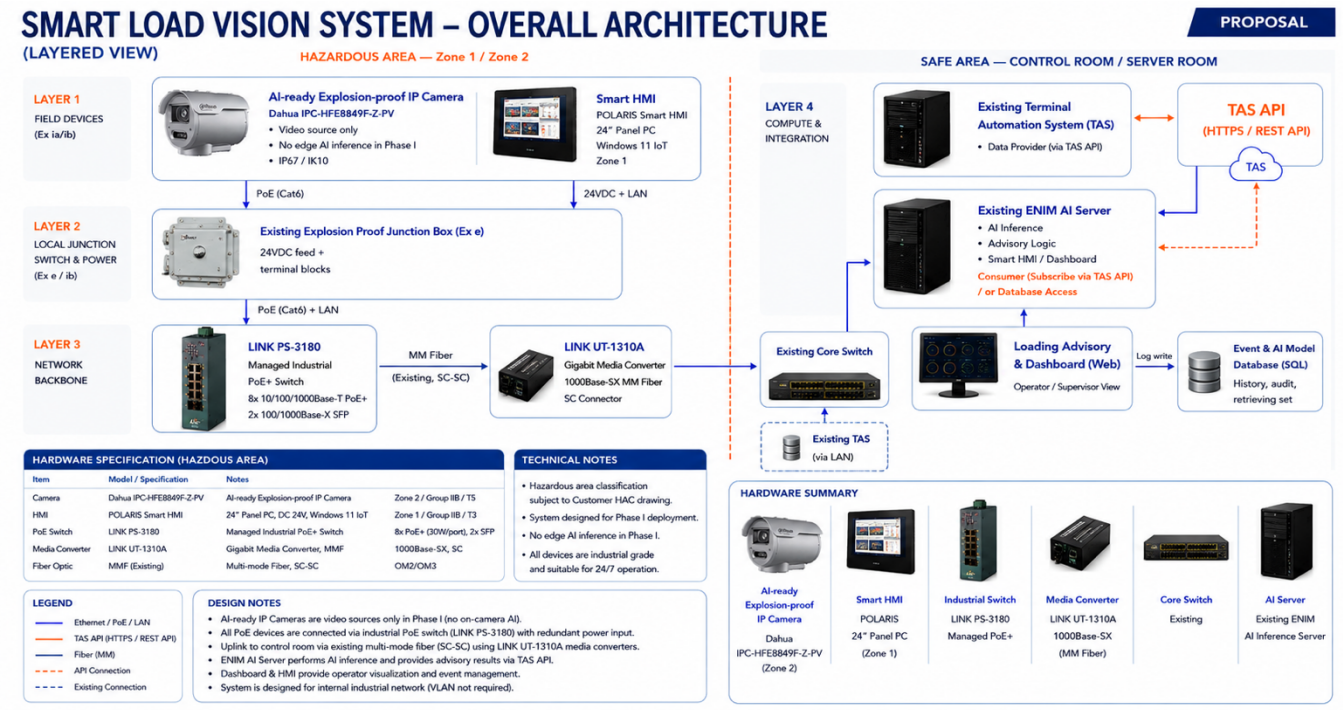


Figure 9.1 — Overall Architecture of the Smart Load Vision System

9.2 Network Architecture

Figure 9.2 illustrates the network connectivity between the AI camera, Smart HMI, Existing ENIM AI Server, and Existing TAS over the Customer's existing terminal LAN. The Existing TAS core is accessed only via TAS API (read-only). Indicative IP addressing is provided in Table 13-1 (Section 13).

Figure 9.2 — Network Connectivity Diagram
Smart Load Vision System — Existing Terminal LAN (no new network segmentation)

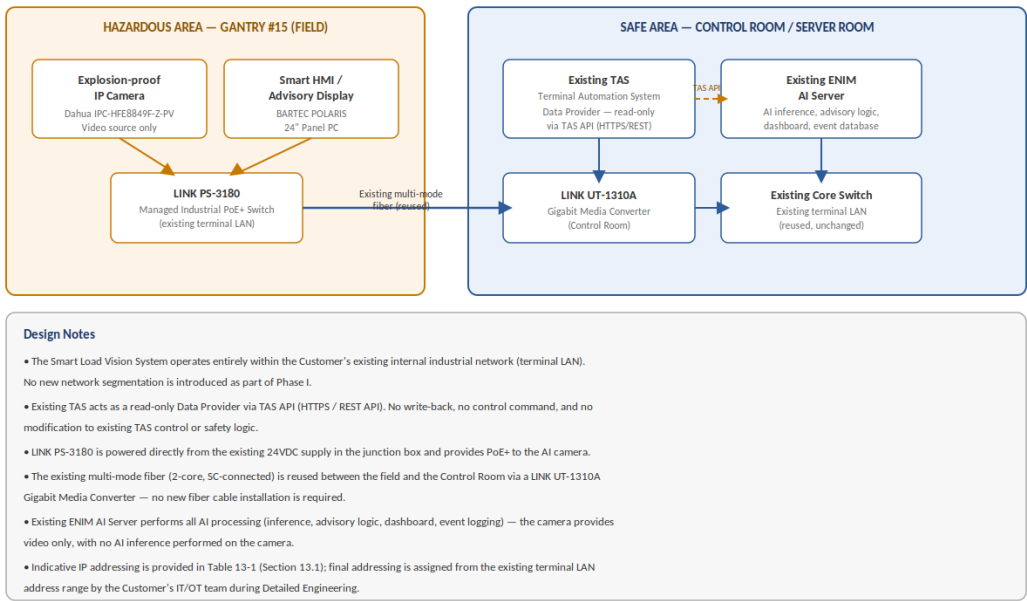


Figure 9.2 — Network Connectivity Diagram

9.3 Data Flow

Figure 9.3 traces a single detection event from image acquisition on the Existing ENIM AI Server through to operator/driver advisory display and long-term logging, including the comparison of the TAS API-provided recommended loading pattern against the AI-detected visual sequence, and the model-retraining feedback loop that is central to the PoC's continuous-improvement objective (Section 5).

Figure 9.3 — Data Flow: Image Acquisition to Operator Advisory

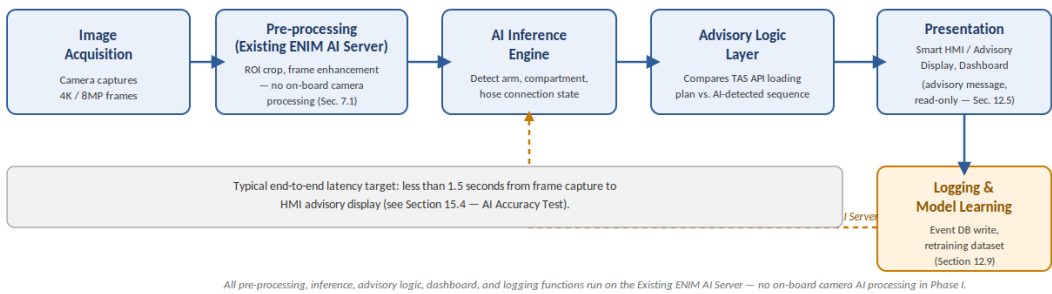


Figure 9.3 — End-to-End Data Flow

9.4 Hardware Connectivity

Figure 9.4 presents the physical wiring schematic between the field-mounted camera and HMI, the existing explosion proof junction box, and the new/existing network and power equipment — including the LINK PS-3180 providing field LAN and PoE to the camera, and the LINK UT-1310A at the Control Room side — together with the recommended cable containment methods (Section 11).

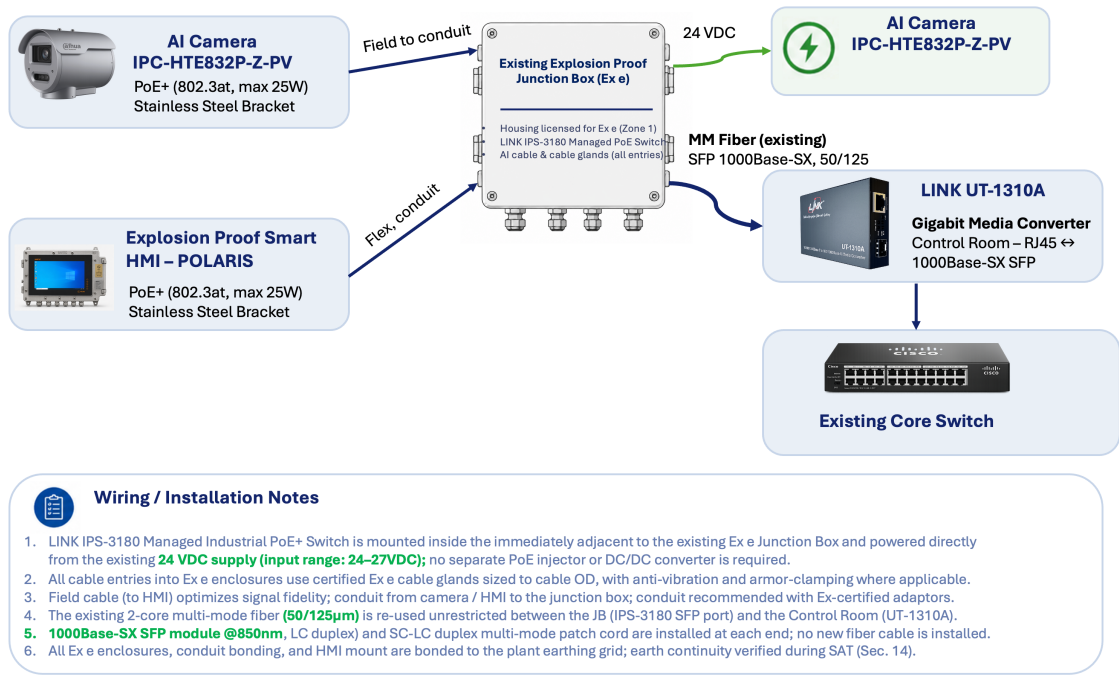


Figure 9.4 — Hardware Connectivity and Field Wiring Schematic